

Assignment 1

Probability and Random Processes

Monsoon 2026

Instructions

- **Deadline:** 26th August 2026
 - Submit handwritten solutions on moodle
 - All questions are compulsory
 - Discussions are encouraged, but copying and plagiarism will be penalised.
 - State your assumptions clearly (*if any*)
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Question 1.

You are given three coins:

- One coin has heads on both faces.
- One coin has tails on both faces.
- One coin has heads on one face and tails on the other.

A coin is selected uniformly at random and tossed, resulting in heads. What is the probability that the opposite face of the coin is tails, given this outcome?

Question 2.

If the probability of observing at least one car on a highway during any 20-minute time interval is $\frac{609}{625}$, then what is the probability of observing at least one car during any 5-minute time interval?

Assume that the probability of seeing a car at any moment is uniform (constant) for the entire 20 minutes.

Question 3.

A certain disease affects about 1 out of 10,000 people. There is a test to check whether the person has the disease. The test is quite accurate. In particular, we know that:

- (a) the probability that the test result is positive (suggesting the person has the disease), given that the person does not have the disease, is only 2 percent;

- (b) the probability that the test result is negative (suggesting the person does not have the disease), given that the person has the disease, is only 1 percent.

A random person gets tested for the disease and the result comes back positive. What is the probability that the person has the disease?

Question 4.

Alice searches for her term paper in her filing cabinet, which has several drawers. She knows that she left her term paper in drawer j with probability $p_j > 0$. The drawers are so messy that even if she correctly guesses that the term paper is in drawer i , the probability that she finds it is only d_i .

Alice searches in a particular drawer, say drawer i , but the search is unsuccessful. Conditioned on this event, find the probability that her paper is in drawer j in terms of p_i , p_j , d_i , and d_j .

Question 5.

Let N be a non-negative integer-valued random variable. Show that

$$\mathbb{E}[N] = \sum_{i=1}^{\infty} P(N \geq i).$$

Question 6.

Give an example of a random variable X such that

$$\mathbb{E}\left[\frac{1}{X}\right] = \frac{1}{\mathbb{E}[X]}.$$

Question 7.

The probability mass function of an infinite discrete distribution is given by $P(X = j) = \frac{1}{2^j}$, $j = 1, 2, \dots, \infty$.

- (a) Verify that the total probability is 1.
- (b) Find the mean and variance of the distribution.
- (c) Find $P(X \text{ is even})$.
- (d) Find $P(X \leq 5)$.
- (e) Find $P(X \text{ is divisible by } 3)$.

Question 8.

Show that, if X is a binomial or Poisson random variable, then the probability mass function (PMF) P_X has the property

$$P_X(k-1)P_X(k+1) \leq P_X(k)^2.$$

Also, give an example of a PMF P_X such that $P_X(k)^2 = P_X(k-1)P_X(k+1)$.

Question 9.

A rifleman can achieve a “marksman” award if he passes a test. He is allowed to fire six shots at a target’s bull’s eye. If he hits the bull’s eye with at least five of his six shots, he wins a set. He becomes a marksman only if he can repeat the feat three times straight, that is, if he can win three straight sets.

If his probability is 0.8 of hitting a bull’s eye on any one shot, find the probabilities of his:

- (a) winning a set, and
- (b) becoming a marksman.

Question 10.

Suppose you’re on the show now, and you’re given the choice of 3 doors. Behind one door is a car; behind the other two, goats. You don’t know ahead of time what is behind each of the doors.

You pick one of the doors and announce it. As soon as you pick the door, Monty opens one of the other two doors that he knows has a goat behind it. Then he gives you the option to either keep your original choice or switch to the third door.

Should you switch? What is the probability of winning a car if you switch?

Question 11.

Two players take turns rolling two fair six-sided dice. Player A goes first, followed by player B. If player A rolls a sum of 6, A is the winner. If player B rolls a sum of 7, B is the winner. If neither player rolls their desired value, the game continues until someone wins.

What is the probability that player A wins?

Question 12.

There is one amoeba in a pond. After every minute the amoeba may die, stay the same, split into two, or split into three with equal probability. All its offspring, if it has any, will behave the same (and independently of other amoebas).

What is the probability that the amoeba population will die out?

Question 13.

You are handed two envelopes, and you know that each contains a positive integer dollar amount and that the two amounts are different. The values of these two amounts are modeled as constants that are unknown. Without knowing what the amounts are, you select at random one of the two envelopes, and after looking at the amount inside, you may switch envelopes if you wish.

A friend claims that the following strategy will increase above $\frac{1}{2}$ your probability of ending up with the envelope with the larger amount:

- Toss a fair coin repeatedly.
- Let X be equal to $\frac{1}{2}$ plus the number of tosses required to obtain heads for the first time.
- Switch if the amount in the envelope you selected is less than the value of X .

Is your friend correct? Justify your answer.

Question 14.

A man is saving up to buy a new car at a cost of Rs. 2000000. He starts with Rs. 200, and tries to win the remaining amount by the following gamble with his bank manager.

He tosses a fair coin repeatedly; if it comes up heads, then the manager pays him one rupee, but if it comes up tails, then he pays the manager one rupee. He plays this game repeatedly until one of two events occurs: either he runs out of money and is bankrupt or he wins enough to buy the car.

What is the probability that he is ultimately bankrupt?

Question 15.

- (a) Prove, or disprove via counterexample: If A is independent of B , and A is independent of C , then A is independent of $B \cup C$.
- (b) For any two events A and B , show that

$$P(A \mid A \cup B) \geq P(A \mid B).$$